

Behavioural Cost and COVID-Safe Behaviour Adherence Over Time in Australia

Final Report

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1 Declaration

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2 Abstract

This study analysed how adherence to 14 COVID-safe behaviours changed over time in Australia and whether behaviours involving greater practical and social burden were less stable. Data was obtained from the Australian component of the Imperial College London–YouGov COVID-19 Behaviour Tracker, containing 53,833 responses across 54 survey waves. The behaviours were classified into low-cost, medium-cost, and high-cost groups. Weighted adherence percentages were calculated across nine six-wave blocks, and logistic regression models were fitted for five behaviours showing the largest increases between qweeks 31–36 and qweeks 37–42.

Low-cost hygiene behaviours stayed high and stable, while medium- and high-cost behaviours showed greater variation over time. Mask wearing recorded the largest increase between the selected periods, followed by avoiding guests and avoiding small, medium-sized, and large gatherings. For all five modelled behaviours, including state-by-period interaction terms significantly improved model fit, showing that the size of the change in behaviour adherence differed across states and territories.

The timing of these changes were consistent with the 2021 Delta outbreak and stronger public-health measures in several states. The findings show that adherence patterns varied with both behavioural burden and state-specific policies.

3 Introduction

Public-health measures depend partially on whether people adapt and maintain the recommended protective behaviours (Ryan et al. 2025, 1–2). During the COVID-19 pandemic, Aus-

traliens were asked to perform behaviours ranging from routine hygiene practices to significant changes in movement and social contact. These behaviours differed in the effort, inconvenience, and disruption they caused, which may have influenced how consistently they were followed (Hinssen and Dohle 2023, secs. 1.1–1.2).

Understanding these differences is important for future public-health planning. Australian research has shown that hygiene behaviours can remain relatively stable while behaviours involving physical distancing change more quickly over time (Ayre et al. 2021, 1). Comparing multiple behaviours can therefore help identify which protective actions remain stable and which respond more strongly to changing outbreak and public-health conditions.

In this report, low-cost behaviours refer to actions requiring little effort and disruption to daily life, while higher-cost behaviours require greater changes to movement, social contact, or normal daily activities.

The following research question is addressed:

How did adherence to COVID-safe behaviours change over time in Australia, and were higher-cost behaviours less stable than lower-cost behaviours?

The project also explores whether the selected change between qweeks 31–36 and qweeks 37–42 differed across Australian states and territories. This report compares 14 protective behaviours grouped by behavioural cost and to distinguish broad national patterns from state-specific differences.

4 Background

4.1 Protective behaviours and adherence

Protective health behaviours reduce the risk of acquiring or transmitting infection. During COVID-19, these included personal hygiene behaviours such as washing hands with soap and water and covering the nose and mouth when coughing or sneezing, as well as mask wearing, avoiding close contact and gatherings, and limiting movement. Their value depends on both effectiveness and continued public adherence.

Adherence can vary between behaviours and across time. Personal hygiene behaviours can generally be performed within an individual’s normal routine, while behaviours such as avoiding gatherings or remaining at home require larger changes to social and daily activities. Previous research has reported associations between mask wearing, social distancing, and hand hygiene, indicating that protective behaviours may occur together (Ryan et al. 2025, 2).

Research has identified several influences on protective behaviour, including demographic characteristics, perceived illness threat, trust in government, social norms, and public-health policy (Ryan et al. 2025, 1–2). The importance of these factors may also change during a pandemic. For example, before a mask mandate, mask wearing may primarily reflect voluntary choice and perceived infection risk; after a mandate, it may also reflect compliance with a legal requirement. Examining several protective behaviours over time can therefore show whether adherence follows a common pattern or changes according to the particular action being considered.

4.2 Behavioural cost and changes over time

Protective behaviours differ in the effort and disruption required to perform them. Some actions, such as washing hands or covering a cough, can be added to an existing routine with relatively

little change. Other behaviours, such as avoiding gatherings, limiting travel, or remaining at home, can restrict social contact, employment, movement, and access to normal activities. These differences can be described as behavioural cost or burden.

Protection Motivation Theory describes these barriers as *response costs*. Response costs can include inconvenience, discomfort, financial expense, or disruption to everyday life. Higher expected response costs may make protective behaviours more difficult to perform even when an individual intends to follow the recommended action (Hinssen and Dohle 2023, secs. 1.1–1.2). In a longitudinal COVID-19 study, perceived costs were negatively associated with subsequently reported physical distancing and hand hygiene, although the corresponding direct effect was not significant for mask wearing (Hinssen and Dohle 2023, sec. 4).

Behavioural burden may also help explain why protective behaviours follow different patterns over time. Australian research conducted during 2020 found that physical-distancing behaviours declined across successive surveys, while hygiene behaviours remained comparatively stable (Ayre et al. 2021, 1). This distinction suggests that behaviours requiring continued changes to movement and social interaction may be less stable than behaviours that can become part of a daily routine.

Behavioural cost was considered one of several influences on adherence. Perceived infection risk, confidence in performing the behaviour, social expectations, and public-health rules may increase adherence despite the burden involved. This report therefore examines both broad cost groups and individual behaviours.

4.3 Risk, policy, and the Australian context

Adherence to protective behaviour can change when people notice a greater risk of infection and when public-health rules become more restrictive. Risk perception, trust in authorities, social expectations, and government policy have all been identified as influences on COVID-19 protective behaviour (Ryan et al. 2025, 1–2). These factors may affect both whether a behaviour is adopted and whether it is maintained over time.

Public-health rules can also change the practical meaning of a behaviour. For example, mask wearing may shift from a voluntary precaution to a legal requirement, while restrictions on gatherings, household visitors, or movement may directly limit opportunities for social contact. Changes in adherence may therefore reflect a combination of voluntary responses to perceived risk and compliance with formal rules.

In Australia, COVID-19 outbreaks and public-health measures differed across states and territories. During the 2021 Delta period, New South Wales, Victoria, and the Australian Capital Territory strengthened stay-at-home orders, movement limits, mandatory mask requirements, and restrictions on gatherings. Western Australia provided a contrasting situation, returning towards pre-lockdown settings in July 2021 with mask and crowd restrictions removed (Australian Bureau of Statistics 2022, section “Delta”; NSW Health 2021, paras. 1–3; State Government of Victoria 2021, paras. 1–7; Barr and Stephen-Smith 2021, paras. 1–6; Perpetch 2021, paras. 1–5).

Because these conditions varied between jurisdictions, a national adherence estimate may combine different state-level patterns. This provides the basis for examining both overall Australian trends and differences between states and territories. However, temporal alignment between policy changes and adherence cannot by itself establish that the policies caused the observed behaviour.

4.4 Logistic regression and model comparison

Logistic regression is used to model an outcome with two possible values, such as adherence (1) and non-adherence (0) (Hosmer, Lemeshow, and Sturdivant 2013). For respondent i , let

$$p_i = P(Y_i = 1 \mid \mathbf{x}_i)$$

represent the probability of adherence given a set of explanatory variables $\mathbf{x}_i$. Because a probability is restricted to values between 0 and 1, logistic regression models the log-odds of the outcome:

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}.$$

The quantity $p_i/(1 - p_i)$ is the odds of adherence. A regression coefficient β_j represents the change in log-odds associated with a one-unit change in x_j , while the other variables in the model are held constant. Exponentiating the coefficient gives an odds ratio:

$$OR_j = \exp(\beta_j).$$

An odds ratio greater than 1 represents higher odds of adherence, an odds ratio below 1 represents lower odds, and an odds ratio equal to 1 represents equal odds. A coefficient p -value tests the null hypothesis that $\beta_j = 0$. A 95% confidence interval describes the range of coefficient values that are reasonably compatible with the fitted model. Confidence intervals for odds ratios are obtained by exponentiating the corresponding coefficient confidence limits.

An interaction term is used when the association between one explanatory variable and the outcome may differ according to another variable. For two variables X and Z , a model containing an interaction can be written as

$$\text{logit}(p_i) = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 (X_i Z_i).$$

The coefficient β_3 measures how the association between X and the outcome differs across values or categories of Z . In this study, the interaction between survey period and state or territory was used to assess whether the period change in adherence differed across jurisdictions.

Logistic-regression coefficients are commonly estimated by maximum likelihood (Hosmer, Lemeshow, and Sturdivant 2013). For independent binary observations, the likelihood is

$$L(\boldsymbol{\beta}) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i},$$

where y_i is the observed binary outcome. The corresponding log-likelihood is

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)].$$

Maximum-likelihood estimation selects the coefficient values that maximise the log-likelihood. For models fitted to the same observations, a larger maximised log-likelihood indicates that the model assigns greater probability to the observed outcomes.

A likelihood-ratio test compares two nested models, where the reduced model is contained within the larger or full model (Hosmer, Lemeshow, and Sturdivant 2013). The likelihood-ratio statistic is

$$LR = 2 (\ell_{\text{full}} - \ell_{\text{reduced}}).$$

Under the null hypothesis that the additional parameters do not improve the model, the statistic is approximately chi-squared distributed. The degrees of freedom equal the number of additional parameters in the full model. A small p-value provides evidence that including those additional parameters improves model fit.

5 Methods

5.1 Overview of the analytical workflow

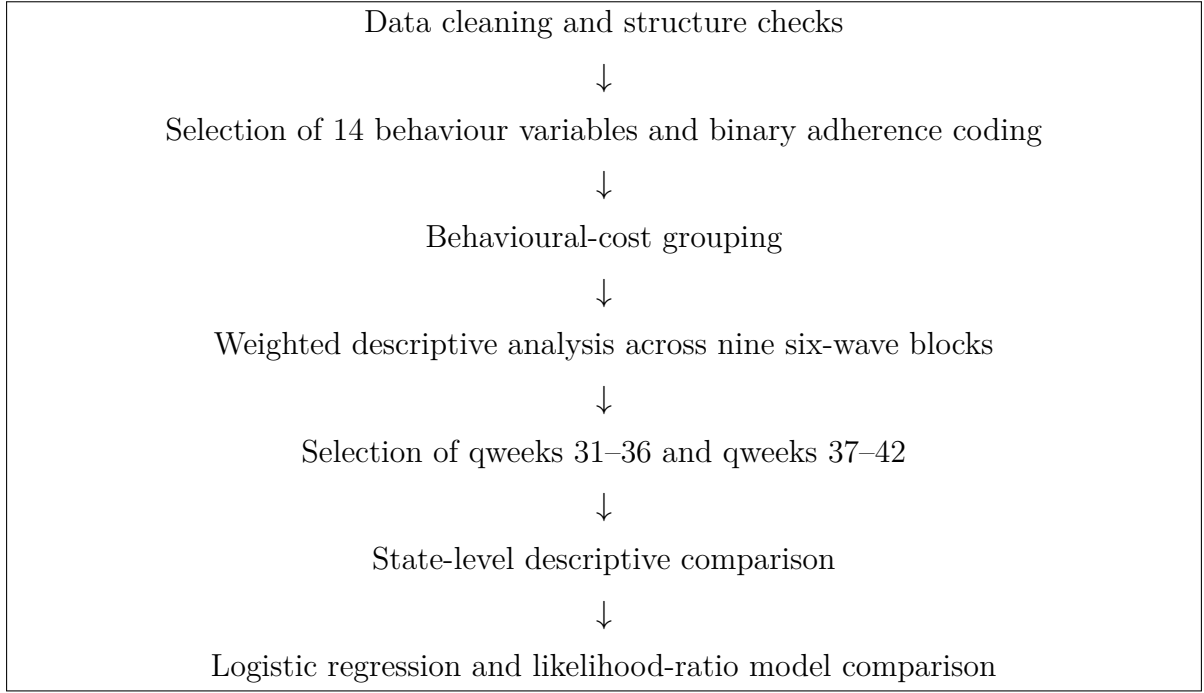
The analysis was carried out in a sequence of linked steps designed to answer the research question of how adherence to COVID-safe behaviours changed over time in Australia and whether higher-cost behaviours were less stable than lower-cost behaviours.

First, the Australian Behaviour Tracker data were cleaned and checked to confirm the dataset structure and the availability of the variables required for the analysis. Second, 14 protective-behaviour items were selected and recoded into separate binary adherence outcomes. Third, the behaviours were classified into low-, medium-, and high-cost groups using a study-specific behavioural-burden framework.

Fourth, weighted descriptive adherence estimates were calculated for individual behaviours and cost groups across nine six-wave blocks in order to identify broad temporal patterns. Fifth, the descriptive results were used to select qweeks 31–36 and qweeks 37–42 as the two adjacent periods showing the clearest simultaneous increase in medium- and high-cost adherence. Sixth, state-level descriptive comparisons were used to examine whether the change between these two periods was similar across jurisdictions.

Finally, logistic regression models were fitted for the five behaviours showing the largest descriptive increases, and likelihood-ratio tests were used to assess whether allowing the period change to differ across states and territories improved model fit. Together, these steps combined descriptive and regression-based analyses to examine both overall adherence patterns and jurisdiction-specific differences.

Analytical workflow



5.2 Data source and study design

This study used data from the Australian component of the Imperial College London–YouGov COVID-19 Behaviour Tracker. The Behaviour Tracker collected repeated cross-sectional survey data on protective behaviours, attitudes, demographics, health, and responses to COVID-19 across multiple countries between 2020 and 2022 (Jones, Imperial College London Big Data Analytical Unit, and YouGov Plc 2020; Ryan et al. 2025, 3).

The working dataset contained 53,833 survey responses collected across 54 survey waves. Each row represented one survey response. All 53,833 responses had a unique RecordNo, with no duplicate values found in the Australian dataset.

The survey was conducted repeatedly over time; however, the dataset did not contain a stable person-level identifier that could be used to link the same respondent across survey waves. The data were analysed as repeated cross-sectional observations. Changes in adherence over time therefore represent differences between the surveyed samples across periods because individual respondents could not be followed across waves.

Variables used in the final analysis included survey wave, survey end date, state or territory, survey weight, and 14 COVID-safe behaviour variables. Demographic variables were retained in the cleaned dataset but were not included in the final regression models.

5.3 Behaviour variables

The analysis included 14 protective-behaviour items from the `i12_health` series of the Behaviour Tracker questionnaire. Each item measured how frequently the respondent reported performing a particular protective behaviour. The original questionnaire wording and variable identifiers were retained where possible to make the selection of behaviours reproducible (Jones, Imperial College London Big Data Analytical Unit, and YouGov Plc 2020, codebook entries `i12_health_1`–`i12_health_8` and `i12_health_11`–`i12_health_16`; Ryan et al. 2025, 4).

The 14 protective-behaviour items were:

- wearing a mask outside the home (i12_health_1);
- washing hands (i12_health_2);
- using hand sanitiser (i12_health_3);
- covering coughs or sneezes (i12_health_4);
- avoiding potentially infected people (i12_health_5);
- avoiding going out (i12_health_6);
- avoiding healthcare settings (i12_health_7);
- avoiding public transport (i12_health_8);
- avoiding guests at home (i12_health_11);
- avoiding small gatherings (i12_health_12);
- avoiding medium-sized gatherings (i12_health_13);
- avoiding large gatherings (i12_health_14);
- avoiding crowded areas (i12_health_15); and
- avoiding shops (i12_health_16).

Each item represented a distinct protective action, so a separate binary adherence outcome was created for each behaviour.

5.4 Binary adherence coding

All 14 behaviours were measured using five ordered response categories: Always, Frequently, Sometimes, Rarely, and Not at all. In the original Behaviour Tracker coding, these responses corresponded to values from 5 for Always to 1 for Not at all (Jones, Imperial College London Big Data Analytical Unit, and YouGov Plc 2020; Ryan et al. 2025, 4).

For this study, each item was dichotomised separately. Always (5) and Frequently (4) responses were classified as adherent (1). Sometimes (3), Rarely (2), and Not at all (1) responses were classified as not adherent (0).

Missing responses were retained as missing and were not assigned to either adherence category. Consequently, the number of observations available for analysis could differ between behaviours.

The same rule was applied separately to all 14 behaviour variables. All 14 binary outcomes were used in the descriptive analysis. The five behaviours selected for the regression analysis were modelled separately as binary dependent variables. The data checks, variable-selection steps, and binary recoding procedure are documented in the Appendix section titled *Data preparation and code references*.

5.5 Behavioural cost grouping

The 14 behaviours were classified into low-, medium-, and high-cost groups using an exploratory behavioural-burden framework developed specifically for this study. The framework has not been externally validated. Behavioural cost was defined as the level of difficulty or disruption involved in maintaining a protective behaviour during normal daily life.

The five scoring dimensions operationalised the response-cost concept discussed in the Background. Each behaviour was assessed according to practical effort, social disruption, lifestyle restriction, emotional burden, and financial burden. These dimensions represented the effort required and the extent to which a behaviour could affect social contact, daily activities, well-being, or finances.

The dimension scores were assigned by the researcher using these definitions and the expected burden imposed by each behaviour. They were not taken from a previously validated behavioural-cost scale.

Each dimension was assigned a score from 1 to 3, where 1 represented low burden, 2 represented moderate burden, and 3 represented high burden. The five scores were added to produce a total behavioural-cost score ranging from 5 to 15. Behaviours with total scores of 5–6 were classified as low cost, scores of 7–9 as medium cost, and scores of 10–15 as high cost.

Table 1 presents the final classification of the 14 behaviours. The complete scoring matrix, including the five individual dimension scores and the total score for each behaviour, is provided in Appendix Table 5.

Table 1: Classification of the 14 protective behaviours by behavioural-cost group.

Behaviour	Behavioural-cost group
Covered coughs or sneezes	Low
Washed hands	Low
Used hand sanitiser	Low
Wore a mask outside the home	Medium
Avoided people with COVID-like symptoms	Medium
Avoided crowded areas	Medium
Avoided public transport	Medium
Avoided having guests at home	Medium
Avoided small gatherings	High
Avoided medium gatherings	High
Avoided large gatherings	High
Avoided going out	High
Avoided shops	High
Avoided healthcare settings	High

The classification was based solely on the five burden dimensions and was completed independently of the observed adherence percentages. This preserved a separation between the classification criteria and the adherence patterns examined in the analysis.

The scoring framework, behaviour-specific rationale, implementation details, and complete numerical scoring matrix are provided in Appendix Section *Behavioural-cost scoring matrix*.

5.6 Survey timing and six-wave blocks

The Behaviour Tracker dataset identified each survey wave using the variable `qweek`. Each `qweek` value represents one survey wave. The intervals between survey waves were not fixed at one calendar week. The 54 `qweek` values were converted into a numeric variable ranging from 1 to 54 so that the survey waves could be placed in their correct chronological order.

The survey end-date variable was used to map each `qweek` to the corresponding calendar period. This allowed the survey-based time periods to be compared with the timing of COVID-19 outbreaks and public-health measures in Australia. A complete mapping between the 54 survey waves and their calendar dates is provided in Appendix Table 6.

For the descriptive analysis, the 54 survey waves were divided into nine consecutive blocks containing six **qweek** values each: qweeks 1–6, 7–12, 13–18, 19–24, 25–30, 31–36, 37–42, 43–48, and 49–54. These groups are referred to as six-wave blocks because each contains six survey waves and may cover a different calendar duration.

The blocks reduced visual clutter while retaining equally sized groups of survey waves. They do not cover equal calendar durations because data collection did not occur at fixed weekly intervals.

After reviewing the nine-block trends, qweeks 31–36 and 37–42 were selected because this adjacent pair showed the clearest simultaneous increase in medium- and high-cost adherence. The period selection was exploratory and was based on the observed descriptive pattern.

Based on the survey end dates, qweeks 31–36 covered 5 May to 27 July 2021, while qweeks 37–42 covered 28 July to 17 October 2021. These two periods were used in the behaviour-level comparisons, state-level comparisons, and logistic regression models. They are referred to throughout the report as the earlier period and the later period, respectively.

The survey-wave mapping, code references, and saved numerical output are provided in Appendix Section *Survey-wave and date mapping*.

5.7 Weighted descriptive adherence estimates

Weighted adherence percentages were calculated using the respondent-level survey-weight variable supplied with the Behaviour Tracker data (Jones, Imperial College London Big Data Analytical Unit, and YouGov Plc 2020). According to the repository documentation, the supplied weighting variable was typically based on respondents’ age, gender, and region. The weights were used as provided in the dataset and were not constructed or recalibrated as part of this study. Their purpose in this analysis was to determine each respondent’s contribution to the weighted adherence estimates.

For behaviour b , the weighted adherence proportion was calculated as

$$\hat{p}_b = \frac{\sum_{i=1}^n w_i Y_{ib}}{\sum_{i=1}^n w_i},$$

where Y_{ib} was the binary adherence outcome for respondent i and behaviour b , and w_i was the respondent’s survey weight. Because adherence was coded as 1 and non-adherence as 0, the weighted mean represented the weighted proportion classified as adherent. This proportion was multiplied by 100 and reported as a percentage. A respondent with a larger survey weight contributed proportionally more to both the numerator and denominator of the estimate. The analytical dataset did not contain the complete survey-design information required to reconstruct the supplied weights independently.

The calculation was performed separately for each of the 14 behaviours across the complete survey period and within each of the nine six-wave blocks. For each estimate, observations were included only when both the relevant adherence outcome and survey weight were available. All 53,833 responses had non-missing, positive survey weights. The handwashing and avoiding-guests variables each contained 1,006 missing responses, involving the same records, while the other 12 behaviour variables had no missing responses. Missing behaviour responses were excluded only from calculations involving that behaviour.

For the descriptive cost-group trends, each respondent’s available binary outcomes were averaged within the low-, medium-, and high-cost groups. These respondent-level group rates were

then summarised using the survey weights within each six-wave block. The grouped estimates were used only to describe broad patterns; individual behaviours remained separate outcomes in the regression analysis.

The same weighted calculation was used to estimate behaviour-specific adherence within each state or territory during qweeks 31–36 and qweeks 37–42. Changes between these periods were calculated in percentage points by subtracting the earlier-period estimate from the later-period estimate.

The corresponding code references and complete descriptive outputs are provided in Appendix Section *Complete descriptive estimates*.

5.8 State-level descriptive comparison

A state-level descriptive comparison was conducted to examine whether the change observed in the national adherence estimates was similar across Australian states and territories. This comparison was included because the timing and extent of COVID-19 outbreaks and public-health measures varied between jurisdictions.

The analysis compared qweeks 31–36 with qweeks 37–42. Based on the survey end dates, these periods covered 5 May to 27 July 2021 and 28 July to 17 October 2021, respectively. For each state or territory represented in the dataset, weighted adherence percentages were calculated separately for the earlier and later periods using the same weighting procedure described above.

For each jurisdiction and period, the respondent-level cost-group rates described above were summarised using survey weights. Change was calculated in percentage points as

$$\text{Change} = \text{later-period adherence} - \text{earlier-period adherence}.$$

A positive value represented an increase in weighted adherence between the two periods, while a negative value represented a decrease.

This descriptive comparison summarised the direction and magnitude of changes across jurisdictions. The regression analysis then examined whether the period change differed across jurisdictions using state-by-period interaction terms.

The unweighted sample sizes, weighted estimates, and percentage-point changes for each jurisdiction are reported in Tables 11 and 12.

5.9 Logistic regression models

The regression analysis examined whether adherence changed between the two selected periods and whether the size of this change differed across states and territories. Five behaviours were modelled separately because they recorded the largest descriptive increases between qweeks 31–36 and qweeks 37–42:

- wearing a face mask outside the home;
- avoiding having guests at home;
- avoiding small social gatherings;
- avoiding medium-sized social gatherings; and
- avoiding large social gatherings.

For each behaviour, the outcome Y_i was coded as 1 for adherence and 0 for non-adherence. The analysis used responses from qweeks 31–42. The period variable was coded as 0 for qweeks 31–36 and 1 for qweeks 37–42.

Two state-based models were fitted separately for each behaviour. Model 2 included survey period and state or territory as main effects:

$$\text{logit}\{P(Y_i = 1)\} = \beta_0 + \beta_1 \text{Period}_i + \beta_2 \text{State}_i.$$

Model 3 included the same main effects together with state-by-period interaction terms:

$$\text{logit}\{P(Y_i = 1)\} = \beta_0 + \beta_1 \text{Period}_i + \beta_2 \text{State}_i + \beta_3 (\text{Period}_i \times \text{State}_i).$$

The state variable was represented by indicator variables for the Australian states and territories. New South Wales was the reference jurisdiction, and qweeks 31–36 were the reference period. In Model 3, β_1 estimated the period change for New South Wales. Each state-by-period interaction coefficient estimated how the period change for another jurisdiction differed from the New South Wales change.

The models included 12,025 complete responses: 5,995 from qweeks 31–36 and 6,030 from qweeks 37–42. The five adherence outcomes, state, period, and survey weight were complete for these responses.

The models were fitted in Python using a binomial generalised linear model with a logit link. The respondent-level survey weights were incorporated into model fitting. HC1 robust covariance estimates were used to calculate coefficient standard errors, 95% confidence intervals, and coefficient p-values (Seabold and Perktold 2010). The corresponding software functions and implementation details are provided in Appendix Section *Complete Model 3 coefficient outputs*.

5.10 Model comparison and statistical reporting

For each of the five behaviours, Model 2 was treated as the reduced model and Model 3 as the full model. The models were compared using a likelihood-ratio test to assess whether adding the state-by-period interaction terms improved model fit.

Model 3 contained seven additional interaction parameters, corresponding to the seven jurisdictions compared with New South Wales. Each likelihood-ratio test therefore used seven degrees of freedom. Test p-values were obtained from the chi-squared distribution (Virtanen et al. 2020). The computational implementation is documented in Appendix Section *Likelihood-ratio test output*.

Model 3 coefficients, odds ratios, 95% confidence intervals, and coefficient p-values were extracted for statistical reporting. The period coefficient for New South Wales and the state-by-period interaction coefficients were used to describe the estimated period differences across jurisdictions. Complete Model 3 coefficient outputs are reported in Appendix Tables 13–17, and the likelihood-ratio test results are reported in Appendix Table 18.

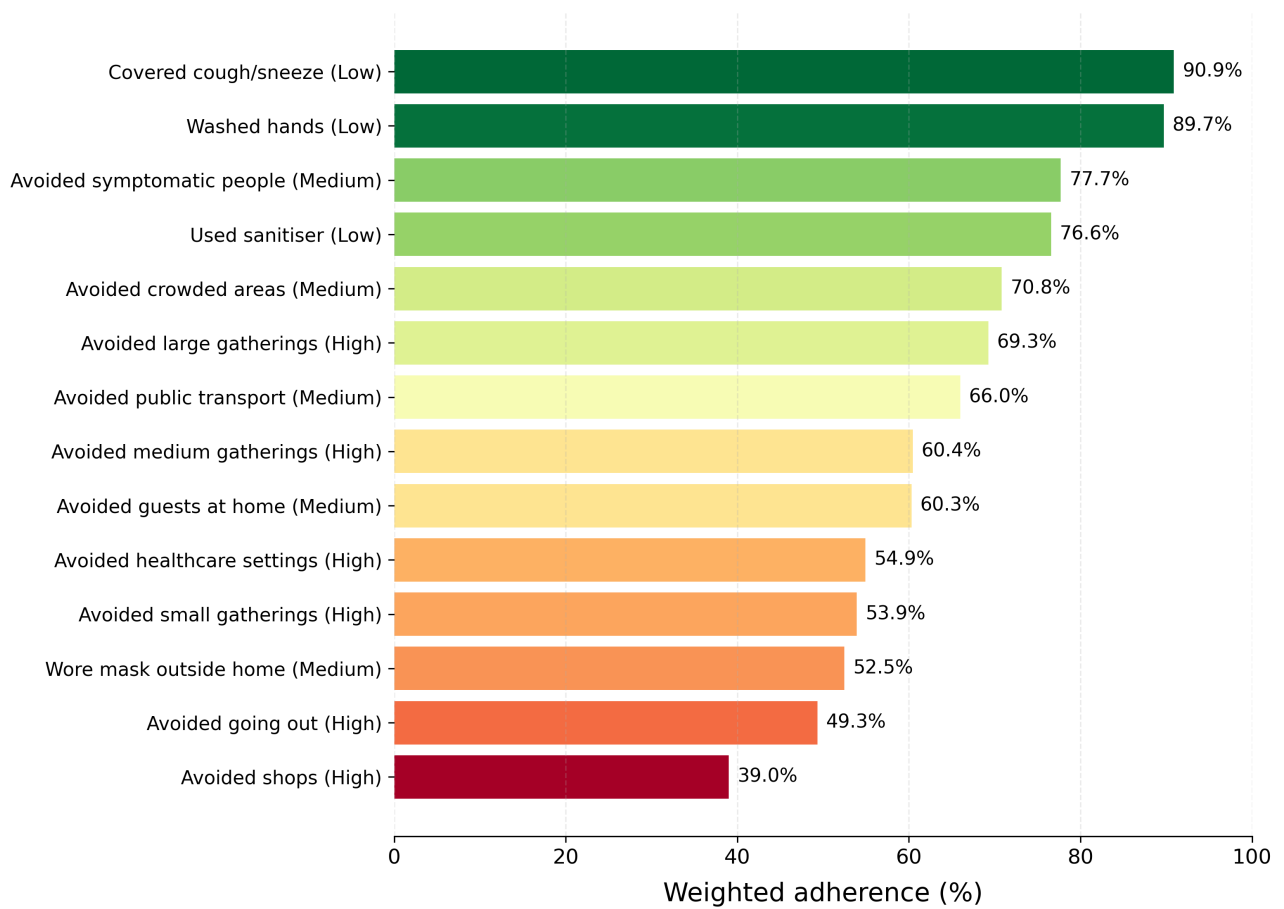


Figure 1: Overall weighted adherence percentage by COVID-safe behaviour.

6 Results

6.1 Overall weighted adherence by behaviour

Figure 1 presents the overall weighted adherence percentages for the 14 protective behaviours. The highest adherence was recorded for covering coughs or sneezes (90.9%) and washing hands (89.7%), while the lowest was recorded for avoiding shops (39.0%), avoiding going out (49.3%), and wearing a face mask outside the home (52.5%).

6.2 Adherence trends across six-wave blocks

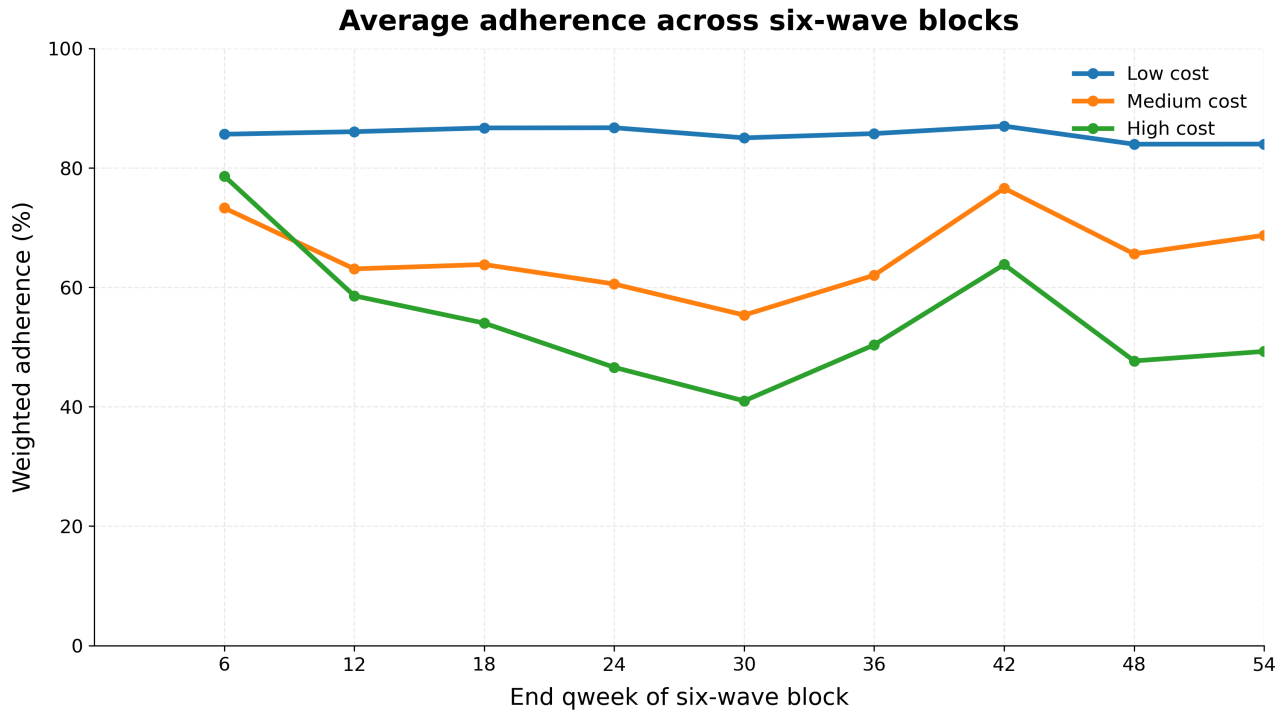


Figure 2: Weighted adherence trends by behavioural-cost group across six-wave blocks.

Figure 2 presents weighted adherence across the nine six-wave blocks. Low-cost adherence ranged from 84.0% to 87.0%, medium-cost adherence ranged from 55.4% to 76.6%, and high-cost adherence ranged from 41.0% to 78.6%. The largest adjacent increase for the medium- and high-cost groups occurred between qweeks 31–36 and qweeks 37–42.

6.2.1 Individual medium-cost behaviour trends

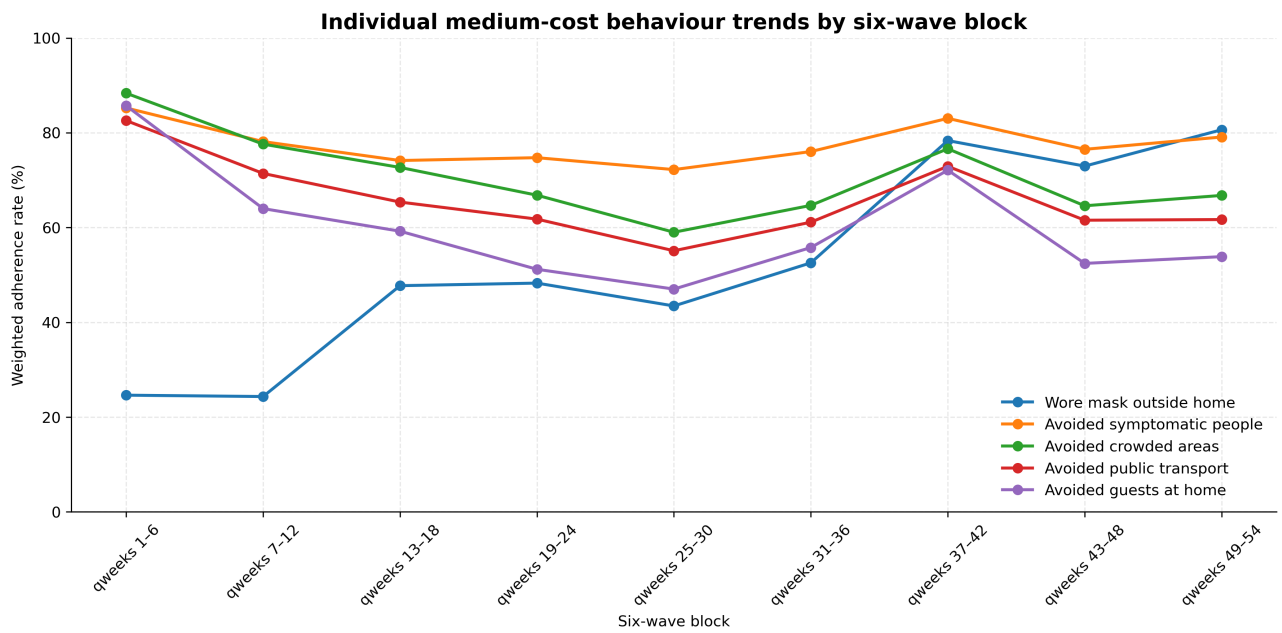


Figure 3: Weighted adherence trends for medium-cost protective behaviours across six-wave blocks.

Figure 3 presents weighted adherence for the five medium-cost behaviours. Between qweeks 31–36 and qweeks 37–42, mask wearing increased from 52.5% to 78.3%, avoiding guests from 55.8% to 72.1%, avoiding crowded areas from 64.7% to 76.6%, avoiding public transport from 61.1% to 72.9%, and avoiding people with COVID-like symptoms from 76.0% to 83.0%.

6.2.2 Individual high-cost behaviour trends

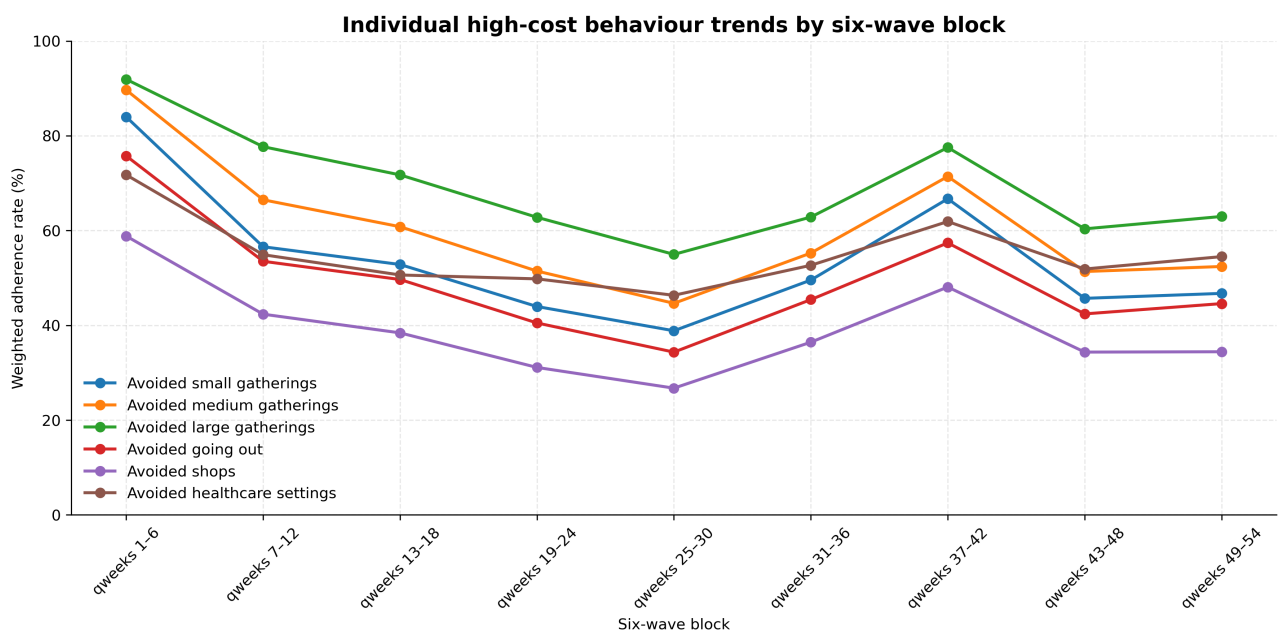


Figure 4: Weighted adherence trends for high-cost protective behaviours across six-wave blocks.

Figure 4 presents weighted adherence for the six high-cost behaviours. Between qweeks 31–36 and qweeks 37–42, adherence increased from 49.5% to 66.7% for avoiding small gatherings, 55.3% to 71.4% for avoiding medium-sized gatherings, 62.8% to 77.5% for avoiding large gatherings, 45.4% to 57.4% for avoiding going out, 36.4% to 48.1% for avoiding shops, and 52.7% to 61.9% for avoiding healthcare settings.

6.3 Behaviour-level change between selected periods

Table 2: Change in weighted adherence between qweeks 31–36 and qweeks 37–42 by behaviour.

Behaviour	Cost group	Earlier (%)	Later (%)	Change (pp)
Wore mask outside home	Medium	52.5	78.3	25.8
Avoided small gatherings	High	49.5	66.7	17.2
Avoided guests at home	Medium	55.8	72.1	16.3
Avoided medium gatherings	High	55.3	71.4	16.2
Avoided large gatherings	High	62.8	77.5	14.7
Avoided going out	High	45.4	57.4	12.0
Avoided crowded areas	Medium	64.7	76.6	11.9
Avoided public transport	Medium	61.1	72.9	11.8
Avoided shops	High	36.4	48.1	11.6
Avoided healthcare settings	High	52.7	61.9	9.2
Avoided symptomatic people	Medium	76.0	83.0	7.0
Covered cough/sneeze	Low	91.0	92.7	1.7
Washed hands	Low	89.5	90.6	1.1
Used sanitiser	Low	76.8	77.7	0.9

Table 2 reports the change between qweeks 31–36 and qweeks 37–42 for all 14 behaviours. Mask wearing recorded the largest increase at 25.8 percentage points, followed by avoiding small gatherings at 17.2 percentage points, avoiding guests at home at 16.3 percentage points, avoiding medium-sized gatherings at 16.2 percentage points, and avoiding large gatherings at 14.7 percentage points. The three low-cost behaviours changed by between 0.9 and 1.7 percentage points.

6.4 State-level descriptive changes

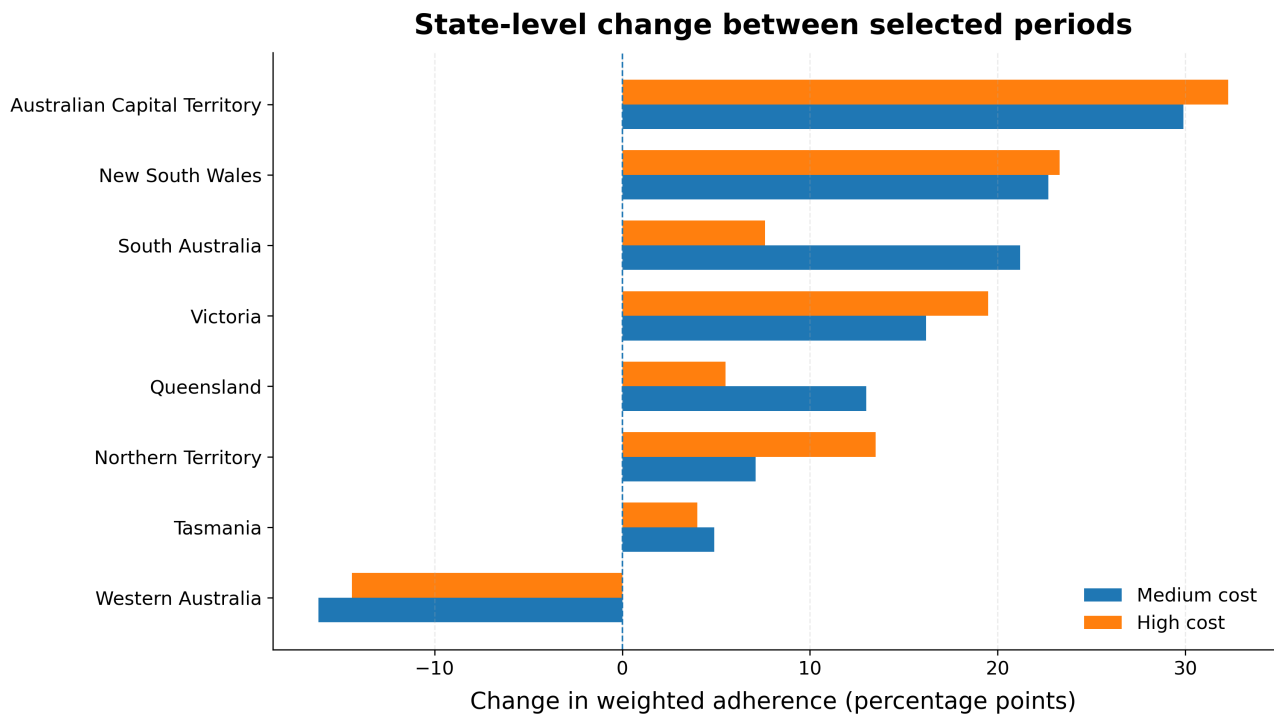


Figure 5: Change in weighted adherence between qweeks 31–36 and qweeks 37–42 by state or territory for the medium- and high-cost groups.

Figure 5 presents the change in medium- and high-cost adherence between the selected periods across states and territories. Medium-cost changes ranged from -16.2 to 29.9 percentage points, while high-cost changes ranged from -14.4 to 32.4 percentage points. Low-cost changes are reported in Appendix Table 12.

6.5 Logistic regression results

Table 3: Later-period coefficients and odds ratios for New South Wales from the state-by-period interaction models.

Behaviour	Period coefficient	Odds ratio	95% confidence interval	P-value
Wearing a face mask outside the home	2.2449	9.44	7.75–11.50	<0.001
Avoiding guests at home	1.5462	4.69	3.97–5.54	<0.001
Avoiding small gatherings	1.4214	4.14	3.56–4.83	<0.001
Avoiding medium-sized gatherings	1.4860	4.42	3.75–5.21	<0.001
Avoiding large gatherings	1.6567	5.24	4.35–6.32	<0.001

Table 3 reports the later-period coefficients and odds ratios for New South Wales from the five state-by-period interaction models. All five period coefficients were positive and statistically significant at $p < 0.001$. The corresponding odds ratios ranged from 4.14 for avoiding small gatherings to 9.44 for mask wearing. Complete state main effects and interaction coefficients are reported in Appendix Tables 13–17.

6.6 Model comparison results

Table 4: Likelihood-ratio tests comparing the period-plus-state model (Model 2) with the state-by-period interaction model (Model 3).

Behaviour	LR statistic	Degrees of freedom	P-value
Mask wearing	662.42	7	<0.001
Avoiding guests	338.32	7	<0.001
Avoiding small gatherings	286.52	7	<0.001
Avoiding medium gatherings	338.39	7	<0.001
Avoiding large gatherings	345.28	7	<0.001

Table 4 reports the likelihood-ratio tests comparing Model 2 with Model 3. The likelihood-ratio statistics ranged from 286.52 to 662.42, with seven degrees of freedom for each comparison. All five tests had $p < 0.001$. For each behaviour, the state-by-period interaction model therefore fitted the data better than the period-plus-state model.

7 Discussion

7.1 Behavioural cost and adherence over time

The results provide general support for the idea that higher-cost protective behaviours were less stable over time than low-cost behaviours. Across the nine six-wave blocks, adherence to low-cost hygiene behaviours remained relatively high, ranging from approximately 84% to 87%. Meanwhile, the medium- and high-cost groups showed substantially greater variation across the survey period.

This difference was also visible between qweeks 31–36 and qweeks 37–42. Covering coughs or sneezes, washing hands, and using hand sanitiser increased by only 1.7, 1.1, and 0.9 percentage points, respectively. These behaviours already had relatively high adherence and could generally be performed without changing movement or social contact. Their stability is consistent with earlier Australian research showing that hygiene behaviours remained comparatively stable while physical-distancing behaviours fluctuated over time (Ayre et al. 2021, 1).

However, behavioural cost did not completely shape adherence. Some medium- and high-cost behaviours increased sharply during the later period. This indicates that the burden of a behaviour can be outweighed by other influences, including outbreak severity, perceived infection risk, social expectations, and public-health rules. This is consistent with evidence that response costs do not affect all protective behaviours in the same way (Hinssen and Dohle 2023, sec. 4). Behavioural cost should therefore be understood as one influence on adherence than as a complete explanation of behaviour.

7.2 Behaviour-specific patterns and public-health context

Mask wearing showed the largest increase between the selected periods, rising by 25.8 percentage points. The next largest changes were seen for avoiding small gatherings at 17.2 percentage

points, avoiding guests at home at 16.3 percentage points, avoiding medium-sized gatherings at 16.2 percentage points, and avoiding large gatherings at 14.7 percentage points.

Mask wearing differed from many other behaviours because it could be directly required in particular settings. During the later period, mask requirements were introduced or strengthened in New South Wales, Victoria, and the Australian Capital Territory (NSW Health 2021, paras. 1–3; State Government of Victoria 2021, paras. 1–7; Barr and Stephen-Smith 2021, paras. 1–6). This provides a plausible explanation for why mask adherence increased more than the other behaviours. Australian research has also shown that mask mandates altered important determinants of adherence to protective health behaviours (Ryan et al. 2025, 1).

Avoiding guests and gatherings also increased substantially. Restrictions introduced in several jurisdictions limited household visitors, private and public gatherings, and permitted reasons for leaving home (State Government of Victoria 2021, paras. 1–7; Barr and Stephen-Smith 2021, paras. 1–6; ABC News 2021, paras. 1–12). These rules directly corresponded to the behaviours measured in the survey, which may help explain their simultaneous increases.

Changes in other movement-related behaviours were smaller. Avoiding going out, shops, crowded areas, public transport, and healthcare settings required greater changes to normal activities. However, some movement remained necessary because restrictions continued to permit essential shopping, healthcare, care responsibilities, and authorised work (State Government of Victoria 2021, paras. 1–7; Barr and Stephen-Smith 2021, paras. 1–6; ABC News 2021, paras. 1–12). This may explain why adherence to avoiding shops and healthcare settings remained lower than adherence to avoiding gatherings.

These interpretations remain cautious because the selected periods contained several simultaneous changes. Outbreak conditions, restrictions, media coverage, perceived risk, and social expectations changed together. The analysis therefore cannot separate the effect of a particular mandate or restriction from these other influences.

7.3 State-level variation and model interpretation

The direction and magnitude of adherence changes varied across Australian states. During the selected qweek period, medium-cost adherence changes ranged from a decrease of 16.2 percentage points in Western Australia to an increase of 29.9 percentage points in the Australian Capital Territory. High-cost changes ranged from a decrease of 14.4 percentage points in Western Australia to an increase of 32.4 percentage points in the Australian Capital Territory.

New South Wales and Victoria also recorded substantial increases in medium- and high-cost adherence. These jurisdictions experienced major Delta outbreaks and stronger restrictions during the later period. The Australian Capital Territory entered lockdown in August 2021 and introduced mandatory masks, retail restrictions, and limits on household and social contact (Barr and Stephen-Smith 2021, paras. 1–6). These conditions correspond closely to the behaviours that increased.

Western Australia showed the opposite pattern, with decreases in both medium- and high-cost adherence. During part of the selected period, Western Australia was removing mask and crowd restrictions following an absence of new local cases (Perpitch 2021, paras. 1–5). This contrast is consistent with adherence changing differently in jurisdictions experiencing different outbreak and policy conditions.

The regression results provided further evidence that the period change differed across jurisdictions. For all five behaviours, the state-by-period interaction model fitted significantly better than the model containing only period and state main effects, with all likelihood-ratio tests

having $p < 0.001$. This means that allowing each jurisdiction to have a different period change improved the models.

For New South Wales, the estimated later-period odds ratios ranged from 4.14 for avoiding small gatherings to 9.44 for mask wearing. These values represent multiplicative changes in the odds of adherence. For example, the mask odds ratio of 9.44 means that the estimated odds of reporting mask adherence in New South Wales were approximately 9.44 times higher during the later period than during the earlier period.

The interaction terms compared each state's period change with the New South Wales change. A negative interaction coefficient indicated that the increase was smaller than the New South Wales increase, while a positive coefficient indicated a larger change. However, the models identify statistical differences between jurisdictions and they also do not show that particular policies caused those differences.

7.4 Limitations

The Behaviour Tracker data were repeated cross-sectional. Different respondents were surveyed across waves, so the findings represent population-level differences between periods rather than changes within the same individuals.

The comparison between qweeks 31–36 and qweeks 37–42 was selected after examining the descriptive trends. It was therefore exploratory and should not be interpreted as a pre-specified evaluation of a particular policy.

The timing of the changes in adherence overlapped with the Delta outbreak and changes in public-health measures across different states, but the analysis does not establish causation. Outbreak severity, perceived risk, media coverage, social expectations, and other unmeasured factors also have influenced adherence.

The behavioural-cost classification was developed specifically for this project and involved literature based judgement. The burden associated with a behaviour may differ between individuals according to their employment, health, household circumstances, transport access, and financial position.

Adherence was measured through self-reported survey responses. Respondents may have understood the questions differently or reported socially desirable behaviour. Survey weighting improves the population relevance of the estimates but cannot remove every form of sampling or response bias.

Grouping the survey waves into six-wave blocks made the broad trends easier to observe, but it may have concealed shorter changes within individual waves. Estimates for smaller jurisdictions should also be interpreted cautiously because they were based on fewer responses. The state-level percentage-point comparisons were descriptive and did not test whether each jurisdiction's individual change was statistically significant. They also cannot attribute the national adherence pattern causally to particular jurisdictions.

Finally, the regression analysis covered only the five behaviours with the largest descriptive increases and did not adjust for demographic characteristics. The supplied respondent-level weights were used for the descriptive estimates and were implemented as frequency weights in the regression models. The analytical dataset did not include the complete survey-design information required to reconstruct the weights or conduct a full design-based survey analysis. The resulting regression estimates should therefore be interpreted as weighted model-based estimates rather than design-based causal effects.

7.5 Future research

Future research could include detailed state-level measures of outbreak severity, policy timing, and restriction intensity. This would allow the relationship between adherence and particular public-health measures to be examined more directly.

Further models could adjust for demographic characteristics such as age, gender and employment. Alternative behavioural-cost classifications could also be tested to determine whether the conclusions depend on the behavioural cost scoring framework.

Longitudinal data linking the same individuals across survey waves would allow future studies to distinguish changes within individuals from changes in the composition of the surveyed population.

8 Conclusion

Answering the research question, higher-cost COVID-safe behaviours were generally less stable over time than low-cost hygiene behaviours. Low-cost behaviours remained comparatively high and stable, while medium- and high-cost behaviours showed greater variation. However, behavioural cost alone did not explain every behaviour-specific pattern.

The largest increases between qweeks 31–36 and qweeks 37–42 occurred for mask wearing, avoiding guests, and avoiding small, medium-sized, and large gatherings. The regression models also showed that the size of these changes varied across states and territories.

The timing of these increases in adherence coincided with the 2021 Delta outbreak and stronger public-health measures across several states and territories. However, because the analysis used repeated cross-sectional data and the periods were selected exploratorily, it cannot establish that individual policies caused the observed changes. Overall, the findings suggest that the adherence was related to both the burden of the behaviour and the jurisdiction-specific outbreak and policy context.

9 Acknowledgements

I would like to acknowledge the guidance and feedback provided by my project supervisor throughout the development of this project.

The analysis code, report source files, figures, tables, and numerical outputs for this project are available in the following public GitHub repository: <https://github.com/7993909690/Research-Part-B>

OpenAI’s ChatGPT was used as a learning and editing aid to support my understanding of statistical concepts and to provide feedback on clarity, structure, grammar, and wording. I independently reviewed and revised the final report, checked the numerical results against the analysis code and saved outputs, and accept responsibility for the submitted work.

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11 Appendix

11.1 Data preparation and code references

The initial data preparation was carried out in `Data_cleaning.ipynb`. The dataset structure and data-quality checks are contained under “Initial Data Quality Check” and “Final dataset structure check”. Selection and renaming of the analysis variables are documented under “Define important columns for my analysis” and “Rename important columns to usable names”. The creation of the 14 binary adherence outcomes is documented under “Create binary adherence variables for each behaviour”.

The cleaned wide-format dataset produced by these steps was saved as `UGov_Data.csv`. These are repository-relative file names; the complete files are available through the public GitHub repository identified in the Acknowledgements.

11.2 Behavioural-cost scoring matrix

Each behaviour was scored from 1 to 3 across practical effort, social disruption, lifestyle restriction, emotional burden, and financial burden. The five scores were summed to obtain the total burden score. Total scores of 5–6 were classified as low cost, 7–9 as medium cost, and 10–15 as high cost. Table 5 presents the complete scoring matrix used in the analysis.

Code and source files: The scoring rationale is documented in `Classification of Protection Behaviours.docx`. The grouping used in the analysis is implemented in `EDA.ipynb`, under “Behaviour columns in wide format”. The table below is generated from `graphs and tables/appendix_A1_cost_scoring.csv`.

Table 5: Behavioural-cost scoring matrix used to classify the 14 protective behaviours.

Behaviour	Practical	Social	Lifestyle	Emotional	Financial	Total	Group
Covering nose/mouth when sneezing or coughing	1	1	1	1	1	5	Low
Washing hands with soap and water	1	1	1	1	1	5	Low
Using hand sanitiser	1	1	1	1	2	6	Low
Wore mask outside home	2	2	1	2	1	8	Medium
Avoiding contact with symptomatic people	1	2	1	2	1	7	Medium
Avoiding crowded areas	1	2	2	2	1	8	Medium
Avoiding public transport	2	1	2	2	1	8	Medium
Avoiding having guests at home	1	2	2	2	1	8	Medium
Avoiding small social gatherings	2	2	2	3	1	10	High
Avoiding medium social gatherings	1	3	2	3	1	10	High
Avoiding large social gatherings	1	3	2	3	1	10	High
Avoiding going out in general	2	3	3	3	1	12	High
Avoiding going to shops	2	2	3	2	1	10	High
Avoiding going to healthcare settings	2	1	3	3	1	10	High

11.3 Survey-wave and date mapping

Table 6 reports the observed response-date range, six-wave block, and unweighted number of responses for each `qweek`.

Code and saved output: The numeric survey-wave ordering was created in `Data cleaning.ipynb` during the construction of `week_num`. The observed date ranges were obtained from `response_time`. The table below is generated from `graphs and tables/appendix_A2_qweek_dates.csv`.

Table 6: Observed response-date range and six-wave block for each survey wave.

qweek	Survey start date	Survey end date	Six-wave block	Unweighted n
1	01 Apr 2020	03 Apr 2020	1–6	973
2	07 Apr 2020	09 Apr 2020	1–6	1006
3	15 Apr 2020	20 Apr 2020	1–6	1002
4	21 Apr 2020	24 Apr 2020	1–6	1004
5	30 Apr 2020	04 May 2020	1–6	1007
6	06 May 2020	07 May 2020	1–6	1014
7	15 May 2020	18 May 2020	7–12	1008
8	03 Jun 2020	05 Jun 2020	7–12	1004
9	09 Jun 2020	11 Jun 2020	7–12	1005
10	24 Jun 2020	28 Jun 2020	7–12	1009
11	08 Jul 2020	13 Jul 2020	7–12	1010
12	22 Jul 2020	27 Jul 2020	7–12	1001
13	06 Aug 2020	10 Aug 2020	13–18	1003
14	20 Aug 2020	25 Aug 2020	13–18	1013
15	02 Sep 2020	07 Sep 2020	13–18	1009
16	16 Sep 2020	26 Sep 2020	13–18	1001
17	02 Oct 2020	12 Oct 2020	13–18	1000
18	14 Oct 2020	19 Oct 2020	13–18	1006
19	28 Oct 2020	04 Nov 2020	19–24	999
20	11 Nov 2020	17 Nov 2020	19–24	1001
21	16 Dec 2020	30 Dec 2020	19–24	1002
22	30 Dec 2020	05 Jan 2021	19–24	924
23	13 Jan 2021	26 Jan 2021	19–24	1001
24	27 Jan 2021	08 Feb 2021	19–24	1003
25	10 Feb 2021	23 Feb 2021	25–30	1011
26	24 Feb 2021	05 Mar 2021	25–30	966
27	10 Mar 2021	19 Mar 2021	25–30	944
28	24 Mar 2021	06 Apr 2021	25–30	1004
29	07 Apr 2021	19 Apr 2021	25–30	1004
30	21 Apr 2021	05 May 2021	25–30	1003
31	05 May 2021	18 May 2021	31–36	969
32	19 May 2021	01 Jun 2021	31–36	1001
33	02 Jun 2021	14 Jun 2021	31–36	1002
34	16 Jun 2021	28 Jun 2021	31–36	1009
35	30 Jun 2021	13 Jul 2021	31–36	1006
36	14 Jul 2021	27 Jul 2021	31–36	1008
37	28 Jul 2021	05 Aug 2021	37–42	1001
38	11 Aug 2021	17 Aug 2021	37–42	1001

39	23 Aug 2021	06 Sep 2021	37–42	1005
40	07 Sep 2021	17 Sep 2021	37–42	1011
41	20 Sep 2021	03 Oct 2021	37–42	1006
42	04 Oct 2021	17 Oct 2021	37–42	1006
43	18 Oct 2021	26 Oct 2021	43–48	1007
44	01 Nov 2021	09 Nov 2021	43–48	1002
45	16 Nov 2021	23 Nov 2021	43–48	1021
46	29 Nov 2021	10 Dec 2021	43–48	1006
47	14 Dec 2021	29 Dec 2021	43–48	978
48	27 Dec 2021	05 Jan 2022	43–48	1011
49	10 Jan 2022	24 Jan 2022	49–54	1010
50	24 Jan 2022	01 Feb 2022	49–54	1007
51	07 Feb 2022	16 Feb 2022	49–54	897
52	21 Feb 2022	01 Mar 2022	49–54	930
53	07 Mar 2022	16 Mar 2022	49–54	996
54	21 Mar 2022	28 Mar 2022	49–54	1006

11.4 Complete descriptive estimates

11.4.1 Overall adherence estimates

Code and saved output: These estimates were generated in `EDA.ipynb`, under “Overall behaviour summary and plot”. The table is generated from `graphs and tables/appendix_A3_overall_adherence.csv`.

Table 7: Overall weighted adherence estimates for the 14 protective behaviours.

Behaviour	Cost group	Valid responses	Weighted adherence (%)
Covered coughs or sneezes	Low	53833	90.9
Washed hands	Low	52827	89.7
Avoided symptomatic people	Medium	53833	77.7
Used hand sanitiser	Low	53833	76.6
Avoided crowded areas	Medium	53833	70.8
Avoided large gatherings	High	53833	69.3
Avoided public transport	Medium	53833	66.0
Avoided medium gatherings	High	53833	60.4
Avoided guests at home	Medium	52827	60.3
Avoided healthcare settings	High	53833	54.9
Avoided small gatherings	High	53833	53.9
Wore mask outside home	Medium	53833	52.5
Avoided going out	High	53833	49.3
Avoided shops	High	53833	39.0

11.4.2 Behavioural-cost group trends

Code and saved output: These estimates were generated in `EDA.ipynb`, under “Create equal six-wave blocks”. The table is generated from `graphs and tables/appendix_A4_cost_group_trends.csv`.

Table 8: Weighted adherence estimates by behavioural-cost group and six-wave block.

Six-wave block	Low cost (%)	Medium cost (%)	High cost (%)
1–6	85.7	73.3	78.6
7–12	86.1	63.1	58.6
13–18	86.7	63.8	54.0
19–24	86.7	60.6	46.6
25–30	85.0	55.4	41.0
31–36	85.8	62.0	50.4
37–42	87.0	76.6	63.8
43–48	84.0	65.6	47.7
49–54	84.0	68.7	49.3

11.4.3 Medium-cost behaviour trends

Code and saved output: These estimates were generated in `EDA.ipynb`, under “Individual behaviour trends by six-wave block”. The table is generated from `graphs and tables/appendix_A5_medium_behaviour_trends.csv`.

Table 9: Weighted adherence estimates for medium-cost behaviours by six-wave block.

Six-wave block	Behaviour	Cost group	Weighted adherence (%)
1–6	Wore mask outside home	Medium	24.6
1–6	Avoided symptomatic people	Medium	85.2
1–6	Avoided crowded areas	Medium	88.3
1–6	Avoided public transport	Medium	82.6
1–6	Avoided guests at home	Medium	85.7
7–12	Wore mask outside home	Medium	24.3
7–12	Avoided symptomatic people	Medium	78.1
7–12	Avoided crowded areas	Medium	77.6
7–12	Avoided public transport	Medium	71.4
7–12	Avoided guests at home	Medium	64.0
13–18	Wore mask outside home	Medium	47.7
13–18	Avoided symptomatic people	Medium	74.1
13–18	Avoided crowded areas	Medium	72.7
13–18	Avoided public transport	Medium	65.4
13–18	Avoided guests at home	Medium	59.2
19–24	Wore mask outside home	Medium	48.3
19–24	Avoided symptomatic people	Medium	74.7
19–24	Avoided crowded areas	Medium	66.8
19–24	Avoided public transport	Medium	61.8
19–24	Avoided guests at home	Medium	51.2
25–30	Wore mask outside home	Medium	43.5
25–30	Avoided symptomatic people	Medium	72.2
25–30	Avoided crowded areas	Medium	59.0
25–30	Avoided public transport	Medium	55.1
25–30	Avoided guests at home	Medium	47.0

31–36	Wore mask outside home	Medium	52.5
31–36	Avoided symptomatic people	Medium	76.0
31–36	Avoided crowded areas	Medium	64.7
31–36	Avoided public transport	Medium	61.1
31–36	Avoided guests at home	Medium	55.8
37–42	Wore mask outside home	Medium	78.3
37–42	Avoided symptomatic people	Medium	83.0
37–42	Avoided crowded areas	Medium	76.6
37–42	Avoided public transport	Medium	72.9
37–42	Avoided guests at home	Medium	72.1
43–48	Wore mask outside home	Medium	73.0
43–48	Avoided symptomatic people	Medium	76.5
43–48	Avoided crowded areas	Medium	64.6
43–48	Avoided public transport	Medium	61.5
43–48	Avoided guests at home	Medium	52.4
49–54	Wore mask outside home	Medium	80.6
49–54	Avoided symptomatic people	Medium	79.1
49–54	Avoided crowded areas	Medium	66.8
49–54	Avoided public transport	Medium	61.7
49–54	Avoided guests at home	Medium	53.9

11.4.4 High-cost behaviour trends

Code and saved output: These estimates were generated in `EDA.ipynb`, under “Individual behaviour trends by six-wave block”. The table is generated from `graphs and tables/appendix_A6_high_behaviour_trends.csv`.

Table 10: Weighted adherence estimates for high-cost behaviours by six-wave block.

Six-wave block	Behaviour	Cost group	Weighted adherence (%)
1–6	Avoided small gatherings	High	84.0
1–6	Avoided medium gatherings	High	89.7
1–6	Avoided large gatherings	High	91.9
1–6	Avoided going out	High	75.7
1–6	Avoided shops	High	58.8
1–6	Avoided healthcare settings	High	71.7
7–12	Avoided small gatherings	High	56.6
7–12	Avoided medium gatherings	High	66.5
7–12	Avoided large gatherings	High	77.7
7–12	Avoided going out	High	53.5
7–12	Avoided shops	High	42.4
7–12	Avoided healthcare settings	High	54.9
13–18	Avoided small gatherings	High	52.8
13–18	Avoided medium gatherings	High	60.8
13–18	Avoided large gatherings	High	71.7
13–18	Avoided going out	High	49.7
13–18	Avoided shops	High	38.4
13–18	Avoided healthcare settings	High	50.6

19–24	Avoided small gatherings	High	44.0
19–24	Avoided medium gatherings	High	51.4
19–24	Avoided large gatherings	High	62.8
19–24	Avoided going out	High	40.5
19–24	Avoided shops	High	31.1
19–24	Avoided healthcare settings	High	49.8
25–30	Avoided small gatherings	High	38.9
25–30	Avoided medium gatherings	High	44.6
25–30	Avoided large gatherings	High	55.0
25–30	Avoided going out	High	34.3
25–30	Avoided shops	High	26.7
25–30	Avoided healthcare settings	High	46.3
31–36	Avoided small gatherings	High	49.5
31–36	Avoided medium gatherings	High	55.3
31–36	Avoided large gatherings	High	62.8
31–36	Avoided going out	High	45.4
31–36	Avoided shops	High	36.4
31–36	Avoided healthcare settings	High	52.7
37–42	Avoided small gatherings	High	66.7
37–42	Avoided medium gatherings	High	71.4
37–42	Avoided large gatherings	High	77.5
37–42	Avoided going out	High	57.4
37–42	Avoided shops	High	48.1
37–42	Avoided healthcare settings	High	61.9
43–48	Avoided small gatherings	High	45.7
43–48	Avoided medium gatherings	High	51.4
43–48	Avoided large gatherings	High	60.4
43–48	Avoided going out	High	42.4
43–48	Avoided shops	High	34.4
43–48	Avoided healthcare settings	High	51.9
49–54	Avoided small gatherings	High	46.7
49–54	Avoided medium gatherings	High	52.4
49–54	Avoided large gatherings	High	63.0
49–54	Avoided going out	High	44.6
49–54	Avoided shops	High	34.4
49–54	Avoided healthcare settings	High	54.5

11.4.5 State-level estimates and sample sizes

Code and saved outputs: The state-level estimates and changes were generated in `EDA.ipynb`, under “State-level changes between the selected periods”. The tables are generated from `graphs` and `tables/appendix_A7a_state_period_estimates.csv` and `graphs` and `tables/appendix_A7b_state_changes`

Table 11: Unweighted sample sizes and weighted cost-group adherence estimates by jurisdiction and period.

State or territory	Period	Unweighted n	Low cost (%)	Medium cost (%)	High cost (%)
Australian Capital Territory	qweeks 31–36	87	83.1	54.6	39.0
Australian Capital Territory	qweeks 37–42	99	86.3	84.5	71.4
New South Wales	qweeks 31–36	1802	87.5	66.6	56.0
New South Wales	qweeks 37–42	1864	89.5	89.3	79.3
Northern Territory	qweeks 31–36	46	79.5	47.6	31.2
Northern Territory	qweeks 37–42	41	83.6	54.7	44.7
Queensland	qweeks 31–36	1197	83.2	53.9	43.9
Queensland	qweeks 37–42	1187	85.4	66.9	49.5
South Australia	qweeks 31–36	614	84.3	50.3	41.1
South Australia	qweeks 37–42	596	85.9	71.5	48.6
Tasmania	qweeks 31–36	121	81.1	37.3	27.9
Tasmania	qweeks 37–42	123	81.8	42.1	31.9
Victoria	qweeks 31–36	1535	87.0	70.6	55.6
Victoria	qweeks 37–42	1555	88.9	86.8	75.1
Western Australia	qweeks 31–36	593	85.1	57.6	46.8
Western Australia	qweeks 37–42	565	80.0	41.4	32.4

Table 12: Percentage-point change in weighted cost-group adherence between the two selected periods.

State or territory	Low cost change (pp)	Medium cost change (pp)	High cost change (pp)
Australian Capital Territory	3.2	29.9	32.4
New South Wales	2.0	22.7	23.3
Northern Territory	4.1	7.1	13.5
Queensland	2.2	13.0	5.6
South Australia	1.6	21.2	7.5
Tasmania	0.7	4.8	4.0
Victoria	1.9	16.2	19.5
Western Australia	-5.1	-16.2	-14.4

11.5 Complete Model 3 coefficient outputs

Code and saved outputs: The five models were fitted in `Modelling.ipynb`, within the behaviour-specific Model 3 sections. Models were implemented in Python using `statsmodels.formula.api.glm()` with a binomial family and logit link. Respondent weights were passed through the `freq_weights` argument, and HC1 robust covariance estimates were used when calculating standard errors, confidence intervals, and coefficient p-values. The tables below are generated from `graphs and tables/appendix_A8_model3_mask_wearing.csv` through `graphs and tables/appendix_A12_model3_avoidin`

New South Wales and qweeks 31–36 were the reference categories. State main effects compare each jurisdiction with New South Wales during qweeks 31–36. Interaction terms show how each jurisdiction’s period change differed from the New South Wales period change.

11.5.1 Wearing a face mask outside the home

Table 13: Complete Model 3 coefficient output for mask wearing.

Term	Coefficient	Odds ratio	95% CI	P-value
Intercept (NSW, qweeks 31–36)	0.338	1.402	1.281–1.536	<0.001
ACT vs NSW (qweeks 31–36)	-0.592	0.553	0.370–0.828	0.004
NT vs NSW (qweeks 31–36)	-1.672	0.188	0.098–0.358	<0.001
QLD vs NSW (qweeks 31–36)	-0.935	0.393	0.338–0.456	<0.001
SA vs NSW (qweeks 31–36)	-1.419	0.242	0.191–0.307	<0.001
TAS vs NSW (qweeks 31–36)	-2.850	0.058	0.030–0.112	<0.001
VIC vs NSW (qweeks 31–36)	0.709	2.032	1.758–2.349	<0.001
WA vs NSW (qweeks 31–36)	-0.529	0.589	0.491–0.707	<0.001
Later period: qweeks 37–42 (NSW)	2.245	9.439	7.754–11.491	<0.001
Later period x ACT	-0.703	0.495	0.260–0.944	0.033
Later period x NT	-1.427	0.240	0.102–0.565	0.001
Later period x QLD	-0.677	0.508	0.391–0.661	<0.001
Later period x SA	0.927	2.526	1.655–3.856	<0.001
Later period x TAS	-2.202	0.111	0.043–0.283	<0.001
Later period x VIC	-0.741	0.477	0.354–0.641	<0.001
Later period x WA	-3.528	0.029	0.021–0.041	<0.001

11.5.2 Avoiding guests at home

Table 14: Complete Model 3 coefficient output for avoiding guests at home.

Term	Coefficient	Odds ratio	95% CI	P-value
Intercept (NSW, qweeks 31–36)	0.466	1.594	1.454–1.748	<0.001
ACT vs NSW (qweeks 31–36)	-0.511	0.600	0.402–0.896	0.012
NT vs NSW (qweeks 31–36)	-0.393	0.675	0.398–1.145	0.145
QLD vs NSW (qweeks 31–36)	-0.611	0.543	0.469–0.628	<0.001
SA vs NSW (qweeks 31–36)	-0.711	0.491	0.397–0.608	<0.001
TAS vs NSW (qweeks 31–36)	-1.340	0.262	0.177–0.387	<0.001
VIC vs NSW (qweeks 31–36)	0.084	1.087	0.947–1.248	0.234
WA vs NSW (qweeks 31–36)	-0.388	0.679	0.565–0.815	<0.001
Later period: qweeks 37–42 (NSW)	1.546	4.693	3.975–5.541	<0.001
Later period x ACT	-0.298	0.742	0.397–1.389	0.351
Later period x NT	-1.697	0.183	0.086–0.389	<0.001
Later period x QLD	-1.192	0.304	0.241–0.383	<0.001
Later period x SA	-1.135	0.322	0.234–0.442	<0.001
Later period x TAS	-1.198	0.302	0.175–0.521	<0.001
Later period x VIC	-0.132	0.877	0.685–1.122	0.295
Later period x WA	-2.203	0.110	0.083–0.146	<0.001

11.5.3 Avoiding small gatherings

Table 15: Complete Model 3 coefficient output for avoiding small gatherings.

Term	Coefficient	Odds ratio	95% CI	P-value
Intercept (NSW, qweeks 31–36)	0.260	1.296	1.185–1.419	<0.001
ACT vs NSW (qweeks 31–36)	-0.686	0.504	0.335–0.758	0.001
NT vs NSW (qweeks 31–36)	-0.811	0.444	0.257–0.768	0.004
QLD vs NSW (qweeks 31–36)	-0.622	0.537	0.463–0.621	<0.001
SA vs NSW (qweeks 31–36)	-0.756	0.470	0.378–0.583	<0.001
TAS vs NSW (qweeks 31–36)	-1.457	0.233	0.153–0.354	<0.001
VIC vs NSW (qweeks 31–36)	-0.007	0.993	0.867–1.136	0.913
WA vs NSW (qweeks 31–36)	-0.509	0.601	0.501–0.722	<0.001
Later period: qweeks 37–42 (NSW)	1.421	4.143	3.558–4.824	<0.001
Later period x ACT	-0.107	0.899	0.491–1.645	0.729
Later period x NT	-1.090	0.336	0.156–0.723	0.005
Later period x QLD	-1.091	0.336	0.269–0.420	<0.001
Later period x SA	-1.095	0.335	0.245–0.458	<0.001
Later period x TAS	-1.305	0.271	0.150–0.489	<0.001
Later period x VIC	-0.147	0.863	0.691–1.079	0.197
Later period x WA	-1.981	0.138	0.104–0.182	<0.001

11.5.4 Avoiding medium-sized gatherings

Table 16: Complete Model 3 coefficient output for avoiding medium-sized gatherings.

Term	Coefficient	Odds ratio	95% CI	P-value
Intercept (NSW, qweeks 31–36)	0.500	1.649	1.504–1.808	<0.001
ACT vs NSW (qweeks 31–36)	-0.727	0.484	0.323–0.724	<0.001
NT vs NSW (qweeks 31–36)	-1.052	0.349	0.202–0.604	<0.001
QLD vs NSW (qweeks 31–36)	-0.576	0.562	0.486–0.651	<0.001
SA vs NSW (qweeks 31–36)	-0.702	0.495	0.401–0.613	<0.001
TAS vs NSW (qweeks 31–36)	-1.226	0.294	0.201–0.429	<0.001
VIC vs NSW (qweeks 31–36)	-0.076	0.927	0.808–1.063	0.279
WA vs NSW (qweeks 31–36)	-0.529	0.589	0.491–0.707	<0.001
Later period: qweeks 37–42 (NSW)	1.486	4.419	3.747–5.213	<0.001
Later period x ACT	0.235	1.265	0.655–2.445	0.484
Later period x NT	-1.078	0.340	0.158–0.732	0.006
Later period x QLD	-1.234	0.291	0.231–0.367	<0.001
Later period x SA	-1.202	0.301	0.219–0.412	<0.001
Later period x TAS	-1.194	0.303	0.177–0.518	<0.001
Later period x VIC	-0.053	0.948	0.745–1.208	0.667
Later period x WA	-2.075	0.126	0.095–0.166	<0.001

11.5.5 Avoiding large gatherings

Table 17: Complete Model 3 coefficient output for avoiding large gatherings.

Term	Coefficient	Odds ratio	95% CI	P-value
Intercept (NSW, qweeks 31–36)	0.721	2.057	1.870–2.263	<0.001
ACT vs NSW (qweeks 31–36)	-0.613	0.542	0.362–0.810	0.003
NT vs NSW (qweeks 31–36)	-0.978	0.376	0.221–0.640	<0.001
QLD vs NSW (qweeks 31–36)	-0.431	0.650	0.560–0.755	<0.001
SA vs NSW (qweeks 31–36)	-0.549	0.577	0.466–0.715	<0.001
TAS vs NSW (qweeks 31–36)	-0.923	0.397	0.277–0.570	<0.001
VIC vs NSW (qweeks 31–36)	0.049	1.050	0.910–1.212	0.501
WA vs NSW (qweeks 31–36)	-0.363	0.696	0.577–0.838	<0.001
Later period: qweeks 37–42 (NSW)	1.657	5.242	4.351–6.315	<0.001
Later period x ACT	0.478	1.613	0.732–3.551	0.235
Later period x NT	-1.178	0.308	0.144–0.660	0.002
Later period x QLD	-1.352	0.259	0.202–0.332	<0.001
Later period x SA	-1.245	0.288	0.207–0.402	<0.001
Later period x TAS	-1.532	0.216	0.128–0.365	<0.001
Later period x VIC	-0.267	0.766	0.585–1.002	0.052
Later period x WA	-2.276	0.103	0.077–0.138	<0.001

11.6 Likelihood-ratio test output

Table 18 reports the likelihood-ratio tests comparing Model 2 with Model 3 for the five modelled behaviours.

Code and saved output: The model comparisons were performed in `Modelling.ipynb`, under “Final model-comparison section”. Likelihood-ratio p-values were calculated using `scipy.stats.chi2.sf()`. The table is generated from `graphs and tables/appendix_A13_likelihood_ratio_t`

Table 18: Likelihood-ratio tests comparing Model 2 with Model 3.

Behaviour	LR statistic	Degrees of freedom	P-value
Mask wearing	662.42	7	<0.001
Avoiding guests	338.32	7	<0.001
Avoiding small gatherings	286.52	7	<0.001
Avoiding medium gatherings	338.39	7	<0.001
Avoiding large gatherings	345.28	7	<0.001

The analysis code, report source file, cleaned dataset, figures, tables, and saved numerical outputs are available in the public GitHub repository provided in the Acknowledgements section.